



MILLPLEDGE *VETERINARY*

THE ANIMILL™
FLUID THERAPY BOOK

This booklet has been produced as a guide only and is not intended to replace official training and education. Fluid therapy should only be undertaken when there is a clinical indication to do so and under the supervision and guidance of a veterinary surgeon. Whilst every precaution has been taken in the preparation of this booklet. Millpledge Ltd do not assume any responsibility for errors or omissions. Neither is any liability assumed for damages resulting from the use of the information contained within.

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ANISET™ GIVING SET

**ANISET™
DIAL-A-FLOW**
(20 drops/ml)

:(20-250 ml/hour range)
:Improving your control
:of fluid per hour



**ANISET™ PAEDIATRIC
DIAL-A-FLOW**
(60 drops/ml)
.....
(5-250 ml/hour range)
Providing greater accuracy
at lower flow rates

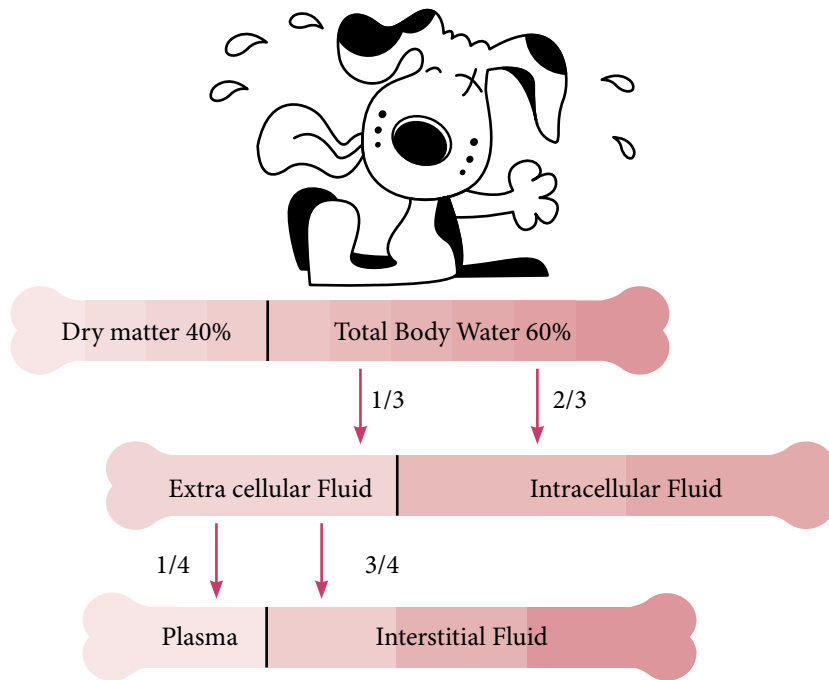
**NON-KINK
TUBING**

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Distribution Of Body Fluids

01



The body contains 60% water; it is distributed throughout the body tissues in different proportions.

The greatest volume is found in the intracellular space, which is inside the cells. The intracellular space equals 40% of the animals' bodyweight and is 2/3 water.

The other 1/3 of the body's water is found outside of the cells in the extra cellular spaces. There are 2 compartments in the extra cellular space. 75% is in the interstitial space, which is also known as the tissue space – the space that surrounds the cells of a given tissue. The water in the interstitial space makes up 15% of the bodyweight. The smaller compartment, 25%, is the intravascular space, which makes up 5% of the bodyweight.

The size of these compartments is controlled within the body by a process called osmosis. This is the movement of water through a semi permeable membrane from an area of low concentration to high concentration until equilibrium is reached. Osmosis is caused by osmotically active particles such as sodium, these particles cannot move through the semi-permeable membrane. Osmosis is responsible for the movement of water from the interstitial fluid into the cells and vice versa.

pH

The pH of blood is dependent on the concentration of hydrogen ions in the blood. The normal pH of blood is 7.35–7.45; this must be maintained for normal cell function to occur. The levels of hydrogen ions in the blood are determined by the amount of bicarbonate and carbon dioxide, which is regulated by the kidneys and respiration. Carbon dioxide is carried in the blood as carbonic acid. A change in pH occurs when a disease process causes a decrease or increase in either hydrogen or bicarbonate.

Acidosis happens when the blood pH is below 7.35. It may be due to a loss of bicarbonate or an increase in hydrogen ions. Acidosis can be either metabolic: being caused by acute renal failure, diabetic ketoacidosis, vomiting, diarrhoea and shock. In addition Acidosis can be respiratory and caused by the animal hypoventilating.

Alkalosis is also broken down into metabolic and respiratory. Alkalosis occurs when the pH of the blood goes above 7.45, and is caused by a rise in bicarbonate or a loss of hydrogen ions. Metabolic alkalosis occurs when an animal has been vomiting, or receives an over infusion of solutions that contain bicarbonate. Respiratory alkalosis is caused by hyperventilation.

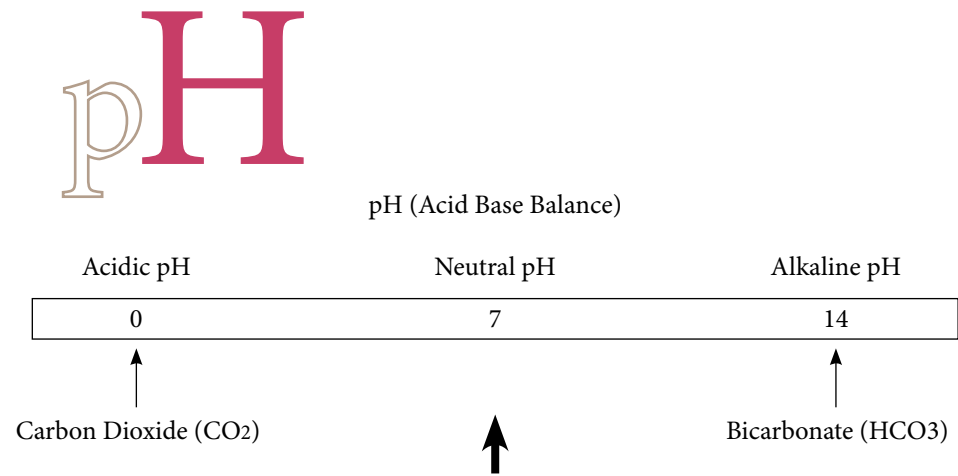
The aim of fluid therapy should be to treat the underlying cause and then in turn the acidosis or alkalosis will also be corrected.

Arterial blood gas analysis will detect acidemia or alkalemia and can help differentiate between metabolic or respiratory.

continued...

Distribution Of Body Fluids

01



CO₂ in the body is converted into Carbonic acid which can quickly be turned into bicarbonate therefore maintaining the neutral pH (equilibrium).

Generally the CO₂ levels and the bicarbonate levels effect each other i.e. if CO₂ level is high the bicarbonate level is low.



	RESPIRATORY ACIDOSIS	METABOLIC ACIDOSIS	METABOLIC ALKALOSIS	RESPIRATORY ALKALOSIS
MEANING	↑ CO₂ (Hypoventilation) Poor ventilation so get increase in CO₂ i.e. pulmonary disease.	↓ BICARBONATE Due to acute renal failure, diabetic ketoacidosis, vomiting, diarrhoea, shock.	↑ BICARBONATE Can be due to bicarbonate overdose or prolonged vomiting.	↓ CO₂ Due to Hyperventilation.
SECONDARY EFFECT	↓ BICARBONATE	↑ CO₂	↓ CO₂	↑ BICARBONATE
MODE OF COMPENSATION	Kidneys will retain bicarbonate therefore increasing the pH.	Lungs ‘blow off’ CO₂ (increase respiratory rate) to decrease the level of CO₂ and therefore increase bicarbonate levels. Kidneys will decrease the amount of bicarbonate they excrete.	Lungs will retain CO₂ (slow the respiratory rate) thereby decreasing the bicarbonate levels. Kidneys will increase the amount of bicarbonate they excrete.	Kidneys will excrete more bicarbonate thereby decreasing the pH.
POSSIBLE TREATMENTS	Need to ensure breathing is adequate i.e. treat the cause of the decreased ventilation.	Some cases require IV administration of sodium bicarbonate solutions.	Treat with antiemetics if persistent vomiting. Possible IV administration of an isotonic sodium chloride solution.	Encourage slowing of respiration rate, or maybe breathe into bag / mask to induce rebreathing of CO₂.

- One or a combination can occur at any given time
- The lungs can make rapid changes to the pH levels, the kidneys can take days to make changes

The processes involved in the body to maintain acid base balance are very complex, the table above demonstrates it in a simplified manner.

Reasons For Fluid Therapy

02

Water intake must be sufficient to meet body needs and prevent dehydration.

The normal healthy animal is able to match the intake of fluids with output of water and electrolytes to maintain normal function. It gets water from two main sources.

- Ingestion – food and fluids
- Metabolism – fats and carbohydrates



Fluids



Food

Animals can lose water through 4 main routes:

- **Kidneys** – This is the only route that the animal can regulate water loss by varying the water content of its urine when water availability is limited.
- **Gastrointestinal tract** – Loss occurs through vomiting, diarrhoea, and through faeces.
- **Respiratory tract** – The animal loses water through breathing and panting, because air is humidified as it passes through the respiratory system.
- **Skin** – Water is lost via sweating.

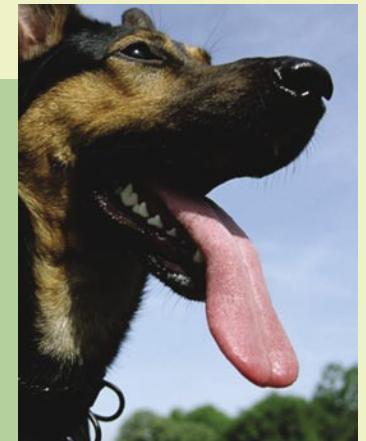
The water lost by the animal via the skin and respiratory system is known as **insensible** loss as the animal cannot regulate these losses, even in times of water deprivation.

The animal loses, via insensible loss, in a 24 hour period.

- Respiratory / skin loss – 20 ml/kg
- Normal faecal loss – 10 – 20 ml/kg
- Normal urinary loss – 20 ml/kg



Sweating



Breathing

Many medical or surgical conditions and injuries can upset the normal water balance of an animal.

- Dehydration due to vomiting or diarrhoea. For the purposes of calculations the amount of fluid lost by the animal per episode of vomiting or diarrhoea is 4 ml/kg bodyweight.
- Significant blood loss or hypovolaemia
- Shock
- Metabolic and endocrine disorders
- Acidosis and alkalosis
- Urinary obstruction
- Renal failure
- Hepatic disease
- Prolonged surgical procedures
- Medication

Animals can also lose water via:

- **Medication** – some types of medications such as diuretics can cause an animal to lose water.
- **Non medical / surgical conditions:** Situations that cause an animal to be stressed i.e. travel, stress, environmental changes, change in diet, could all cause conditions such as diarrhoea, which in turn could affect the water balance of an animal.

Assessing Dehydration

03

It is important to assess how dehydrated an animal is before you commence any fluid therapy. There are several clinical and laboratory methods used to assess how much fluid an animal has lost. We will look at a couple of the more simple ones.

History

Allows you to assess how much fluid the animal has lost. The owner should be asked.

1. Has the pet been eating and drinking recently?
2. Has there been any vomiting or diarrhoea, and how many episodes have there been?
3. Has the animal been passing urine?
4. Has the animal had any abnormal discharge?
5. Have there been any traumatic injuries that might indicate haemorrhage or burns?

Laboratory Analysis

There are some simple lab tests which can help estimate fluid loss.

1. **Packed Cell Volume Test (PCV)** – inexpensive and quite revealing. For each 1% increase in PCV a fluid loss of 10 ml/kg has occurred. This can be unreliable where there is pre-existing anaemia.
2. **Haemoglobin** – dehydration results in an increase in haemoglobin values, but care should be taken when interpreting results from an anaemic animal.
3. **Total Plasma Protein (TPP)** – dehydration causes a rise in values. This can be measured from either a biochemistry test or from your refractometer. Measuring the total plasma protein is done in the same way as measuring the specific gravity of urine.
4. **Blood urea & creatinine** – parameters will rise in the dehydrated patient but will also be high in an animal with renal disease.
5. **The specific gravity of urine** – can also be an indicator of dehydration. If urine is highly concentrated this can be a sign of dehydration. (See page 44 for normal urine SG ranges)

Clinical Signs

These are useful but not always accurate when used alone.

% Dehydration Values	Clinical Signs Shown
< 5 %	No clinical signs
5 – 8 %	Slight skin tenting Slight increase in CRT Tacky MM 3rd eyelid visible
8 – 10 %	Obvious skin Tenting Sunken eyes Prolonged CRT
10 – 12 %	Skin stands in place Oliguria (< 0.5 ml/kg/hr) Shock
> 12 %	Progressive shock Coma Death



Skin tenting



Packed Cell Volume Test (PCV)

The PCV is the percentage of whole blood that is taken up by the red blood cells. The PCV will fall in patients with anaemia and may be elevated in patients, that are dehydrated.

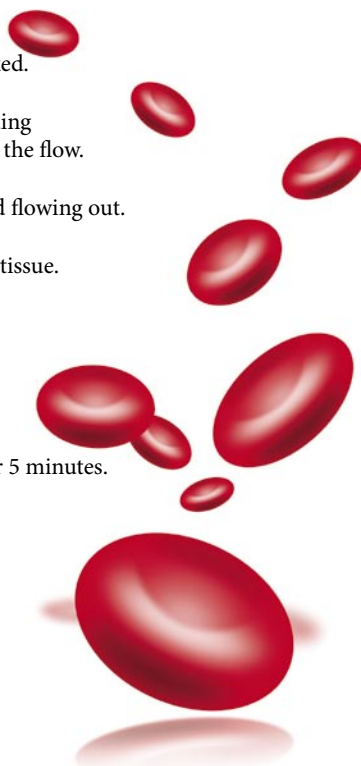
Normal PCV

Species		Average %		Range
Cat	→	35%	→	24 – 45%
Dog	→	45%	→	37 – 55%
Rabbit	→	40%	→	33 – 50%

The PCV is measured by centrifuging a capillary tube of whole blood using a centrifuge then calculating what percentage of the blood volume is red blood cells. PCV's are usually measured using blood collected by venous puncture and collected into EDTA anti-coagulant. Heparinized capillary tubes are available so the sample can be collected directly into the capillary tube via a small nick in the animal's skin. A microhaematocrit reader or a ruler can be used to measure the cells to determine the PCV.

How to determine the PCV

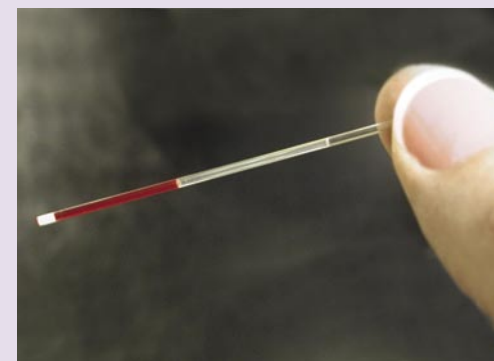
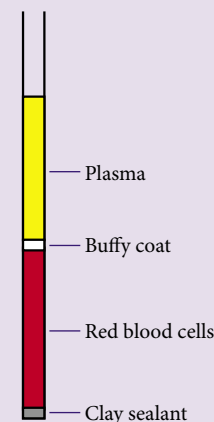
1. Invert the blood tube to ensure that sample is thoroughly mixed.
2. Wearing gloves, three-quarters fill two capillary tubes by holding both the blood tube and capillary tube at an angle to enhance the flow.
3. Place a finger at one end of the capillary tube to prevent blood flowing out.
4. Wipe the capillary tubes clean of excess blood with a piece of tissue.
5. Plug opposite end of the tube with soft clay or haematocil™.
6. Place both capillary tubes on opposite sides of the centrifuge, with the plug against the rubber rim of the centrifuge.
7. Close and lock the centrifuge lid. Centrifuge at 10,000rpm for 5 minutes.
8. Remove the capillary tubes and measure PCV using a microhaematocrit reader or, with a ruler. The sample will be divided into three layers.
 - i) Plasma
 - ii) White blood cells (buffy coat)
 - iii) Red cells



How to measure your PCV with a ruler.

The PCV is calculated by:

Measuring the length of the red cell layer and dividing it by the total length of all 3 layers, this figure is then multiplied by 100 to give the PCV as a percentage.



The above diagram and photo displays how whole blood is separated after being centrifuged.

Combined interpretation of PCV and Total Protein (TP)

PCV	TP	Interpretation of PCV and TP
↑	↑	Dehydration
↑	N or ↓	Splenic contraction, polycythaemia, dehydration + hypoproteinaemia
N	↓	Hyperglobulinaemia, anaemia + dehydration
N	N	Normal, dehydration + anaemia + hypoproteinaemia, acute haemorrhage
↓	N	Anaemia through non-blood loss + normal hydration
↓	↑	Anaemia + dehydration, anaemia + hypoproteinaemia
↓	↓	Blood loss, anaemia + hypoproteinaemia, overhydration

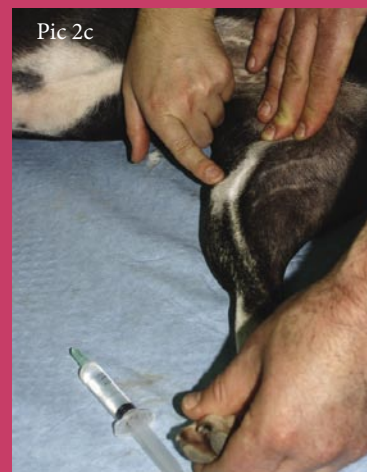
Key: N = Normal

Routes For Fluid Therapy

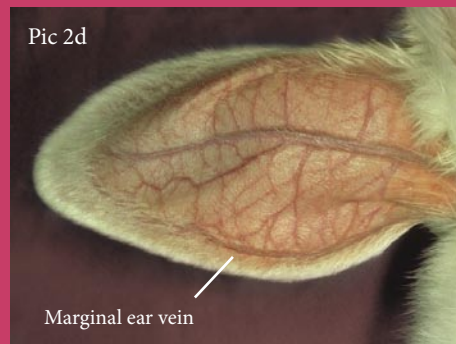
05

The route chosen to administer fluids will depend on the species, and size of the patient, the reason for fluid therapy and the fluid to be administered.

1. **Oral** – Inexpensive to the practice as no specialised equipment is needed. The fluid does not need to be sterile. This method is not recommended for acute or emergency cases and should not be used for cases where the patient is vomiting. Manually administered oral fluids, via syringe, can be highly stressful and could potentially result in aspiration pneumonia.



2. **Peripheral Intravenous** – Cephalic (foreleg, pic 2a), saphenous (Hind leg, lateral pic 2b or medial pic 2c), marginal ear vein (commonly used in rabbits pic 2d) or the jugular vein can be used. Used for all cases of dehydration and shock, where rapid replacement of circulating blood volume and electrolyte adjustment is required. The peripheral route is the only route that is chosen for hypertonic solutions. Surgical preparation of site is required. Cannulae should be changed every 48 – 72 hours, and if possible swap legs / ears.



3. **Subcutaneous** – Only appropriate for mild dehydration. Multiple sites over the shoulders, back and hindquarters can be used. Only use non-irritant and isotonic fluids. Give 10ml / kg, this is absorbed over 6 – 8 hours so not appropriate for emergency or acute conditions. (Pic 3a & 3b)



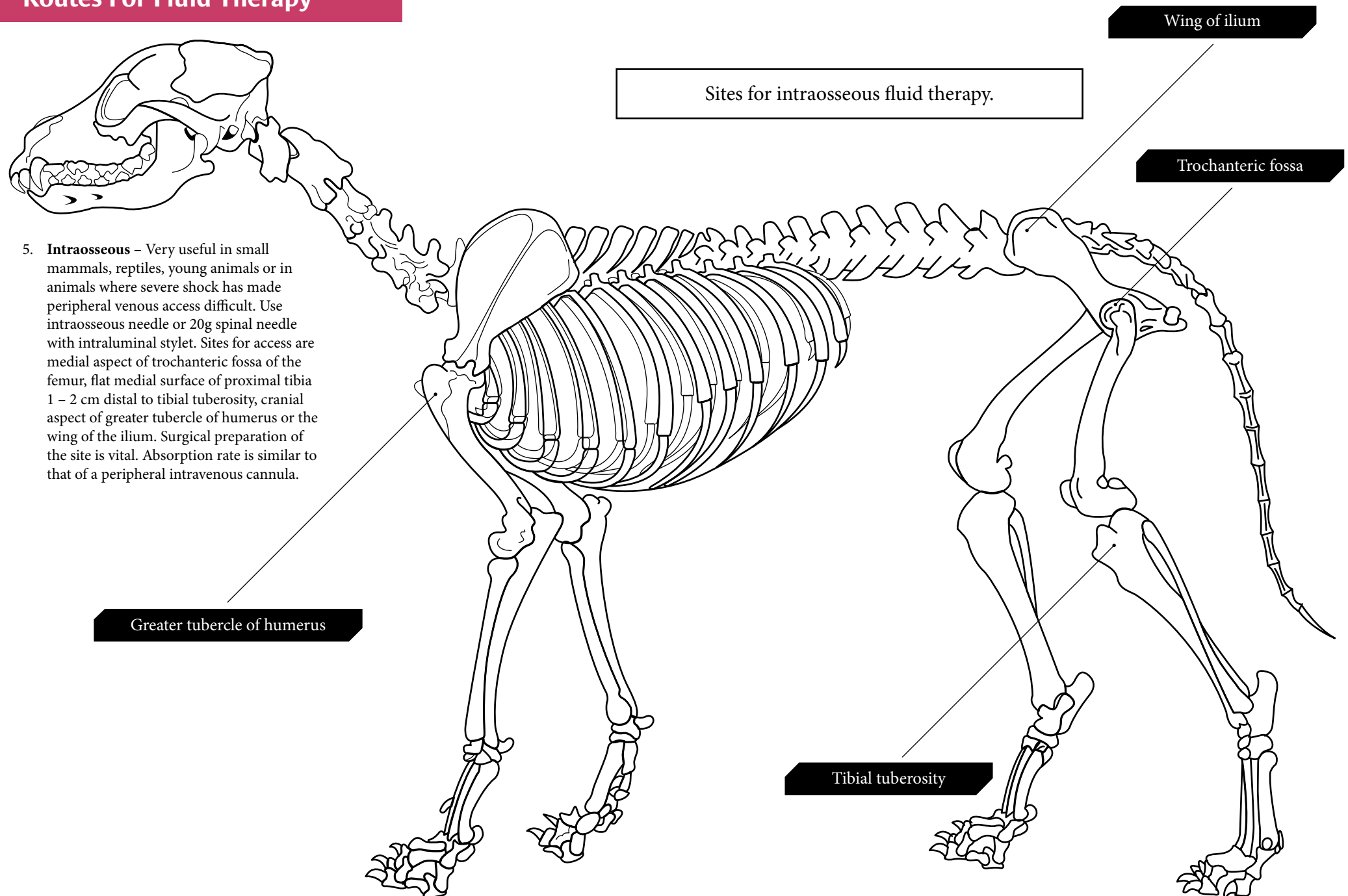
4. **Intraperitoneal** – Fluid can be injected intra-abdominally from the caudal abdomen to the umbilicus. Useful in small mammals and very young animals. Not appropriate for emergency or acute situations. Surgical preparation of the site is required. (Pic 4)



continued...

Routes For Fluid Therapy

5. **Intraosseous** – Very useful in small mammals, reptiles, young animals or in animals where severe shock has made peripheral venous access difficult. Use intraosseous needle or 20g spinal needle with intraluminal stylet. Sites for access are medial aspect of trochanteric fossa of the femur, flat medial surface of proximal tibia 1 – 2 cm distal to tibial tuberosity, cranial aspect of greater tubercle of humerus or the wing of the ilium. Surgical preparation of the site is vital. Absorption rate is similar to that of a peripheral intravenous cannula.



Calculating Fluid Requirements

06

In order to estimate the fluid requirement for an animal, you must first calculate how much the animal has already lost (fluid deficit) and how much the animal needs on a daily basis (maintenance).

We can use the PCV to tell us how much as a percentage the animal is dehydrated and we can also use clinical signs.



REMEMBER

For every 1% increase in PCV or dehydration, the animal has lost 10 ml/kg bodyweight.

Every time the animal vomits or has diarrhoea, we must add 4ml/kg bodyweight to the calculation.

Maintenance fluids

Rabbits	Cats & small dogs	Medium dogs	Large Dogs
0.2 – 4 kg	0 – 10 kg	10 – 30 kg	30 – 50 kg
75 ml/kg/ 24 hours	60 ml/kg/ 24 hours	50 ml/kg/ 24 hours	40 ml/kg/ 24 hours

Example calculation

You are presented with a 30kg dog, which has vomited once. After running a haematology blood test the PCV result is 50%.



How much fluid does the patient require?

1. Work out the fluid deficit:

PCV of 50% (normal PCV is 45%) equals a 5% increase in PCV.

$$\begin{array}{c} 5 \\ \text{\% increase} \\ \text{in PCV} \end{array} \times \begin{array}{c} 10 \\ \text{ml lost per} \\ \text{1\% increase} \end{array} \times \begin{array}{c} 30 \\ \text{kg bodyweight} \end{array} = \begin{array}{c} 1500 \text{ ml} \\ \text{Deficit} \end{array}$$

The animal has vomited once.

$$\begin{array}{c} 4 \\ \text{ml/vomit} \end{array} \times \begin{array}{c} 30 \\ \text{kg bodyweight} \end{array} = \begin{array}{c} 120 \text{ ml} \\ \text{Deficit due to} \\ \text{vomiting} \end{array}$$

So...

$$\begin{array}{c} 120 \text{ ml} \\ \text{Deficit due to} \\ \text{vomiting} \end{array} + \begin{array}{c} 1500 \text{ ml} \\ \text{Deficit due} \\ \text{to \% increase} \\ \text{in PCV} \end{array} = \begin{array}{c} 1620 \text{ ml} \\ \text{Total deficit} \end{array}$$

2. Work out maintenance requirement:



REMEMBER

30kg dog needs 50ml/kg/24 hours

$$\begin{array}{c} 50 \\ \text{ml daily} \\ \text{requirement/kg} \end{array} \times \begin{array}{c} 30 \\ \text{kg bodyweight} \end{array} = \begin{array}{c} 1500 \text{ ml} \\ \text{Over 24 hours.} \\ \text{Maintenance} \\ \text{requirement} \end{array}$$

3. Total Fluid Requirement:

$$\begin{array}{c} 1620 \text{ ml} \\ \text{fluid deficit} \end{array} + \begin{array}{c} 1500 \text{ ml} \\ \text{maintenance} \\ \text{requirement} \end{array} = \begin{array}{c} 3120 \text{ ml} \\ \text{Total required} \\ \text{over 24 hours} \end{array}$$

This fluid should be given over 24 hours unless the animal is showing signs of severe shock and hypovolaemia. In this case the deficit should be replaced over 4 – 6 hours and then the remaining maintenance fluid given over 24 hours.

Calculating The Drip Rate

07

Once you have calculated how much fluid the animal needs you must then calculate the drip rate.

1. Calculate ml per hour:

3120 ml
per daily
requirement

÷

24
hours

=

130 ml
per hour

If you are using an infusion pump or if using an Aniset™ Dial-a-flow (see page 31) this may be as far as you need to take the calculation as most infusion pumps require the mls per hour to be input into the machine.

2. Calculate ml per Minute:

130 ml
per hour

÷

60
minutes

=

2.16 ml
per minute

3. Calculate the drops per minute:

You will need to know how many drops your giving set delivers per ml. Standard Animill™ sets are usually 20 drops/ml and paediatric are usually 60 drops per ml, but check the packaging. An Aniset™ (standard 20 drops/ml giving set) is being used in this case.

2.16 ml
per minute

X

20 drops
Giving set rate
usually 20 or
60 drops

=

43 drops
per minute

4. Calculate drip rate:

60
seconds per
minute

÷

43 drops
Drops per
minute

=

1 drop
every 1.3
seconds

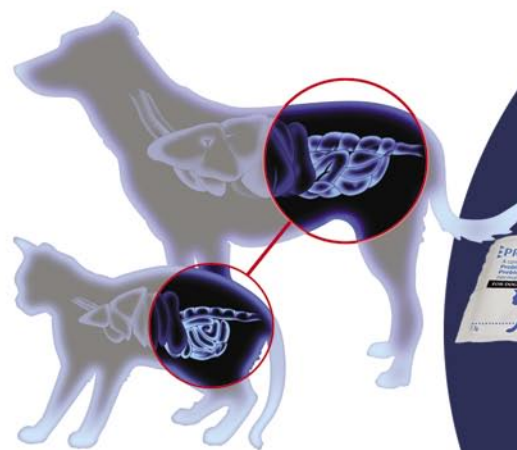
REMEMBER

You must NEVER 'round-up' your final answer. Over infusion can easily occur.

PROMODULATE™

A concentrated source of **Probiotic** & targeted **Prebiotic** to maintain normal gut function

FOR DOGS & CATS



Typically dogs and cats experience excessive loss of fluid with digestive upset, with this the loss of essential electrolytes, Promodulate™ contains a measured source of electrolytes maintaining the isotonic fluid balance and electrolyte level.

Norm-Prozorb™

A unique oral paste presentation of a nutritional supplement as a gut conditioner to assist in the management of digestive tract upsets in dogs, cats and other small animals.



Fluid Therapy Chart

Rabbits

5%

Body weight kg	Deficit ml	Maintenance ml	Drops/min**	ml/hour in 24 hours*
0.2	10	15	1	1
0.4	20	30	2	2
0.6	30	45	3.1	3.1
0.8	40	60	4.1	4.1
1	50	75	5.2	5.2
1.2	60	90	6.2	6.2
1.4	70	105	7.2	7.2
1.6	80	120	8.3	8.3
1.8	90	135	9.3	9.3
2	100	150	10.4	10.4
2.2	110	165	11.4	11.4
2.4	120	180	12.5	12.5
2.6	130	195	13.5	13.5
2.8	140	210	14.5	14.5
3	150	225	15.6	15.6
3.5	175	262.5	18.2	18.2
4	200	300	20.8	20.8

10%

Body weight kg	Deficit ml	Maintenance ml	Drops/min**	ml/hour in 24 hours*
0.2	20	15	1.4	1.4
0.4	40	30	2.9	2.9
0.6	60	45	4.3	4.3
0.8	80	60	5.8	5.8
1	100	75	7.2	7.2
1.2	120	90	8.7	8.7
1.4	140	105	10.2	10.2
1.6	160	120	11.6	11.6
1.8	180	135	13.1	13.1
2	200	150	14.5	14.5
2.2	220	165	16	16
2.4	240	180	17.5	17.5
2.6	260	195	18.9	18.9
2.8	280	210	20.4	20.4
3	300	225	21.8	21.8
3.5	350	262.5	25.5	25.5
4	400	300	29.1	29.1

Cats / Small Dogs

5%

Body weight kg	Deficit ml	Maintenance ml	Drops/min**	ml/hour in 24 hours*
0.5	25	30	2	2
1	50	60	4.5	4.5
2	100	120	9.1	9.1
3	150	180	13.7	13.7
4	200	240	18.3	18.3
5	250	300	22.9	22.9
6	300	360	27.5	27.5
7	350	420	32	32
8	400	480	36.6	36.6
9	450	540	41.2	41.2
10	500	600	45.8	45.8

10%

Body weight kg	Deficit ml	Maintenance ml	Drops/min**	ml/hour in 24 hours*
0.5	50	30	3.3	3.3
1	100	60	6.6	6.6
2	200	120	13.3	13.3
3	300	180	20	20
4	400	240	26.6	26.6
5	500	300	33.3	33.3
6	600	360	40	40
7	700	420	46.6	46.6
8	800	480	53.3	53.3
9	900	540	60	60
10	1000	600	22.2***	66.6

Fluid should be given over 24 hours unless the animal is showing signs of severe shock and hypovolaemia. In this case the deficit should be replaced over 4 – 6 hours and then the remaining maintenance fluid given over 24 hours.

TOTAL AMOUNT TO BE GIVEN

DEFICIT REPLACEMENT =
% dehydration x body weight

MAINTENANCE =

Rabbits 75 ml/kg/24 hours
Cats / Small dogs 60 ml/kg/24hrs
Medium dogs 50 ml/kg/24hrs
Large dogs 40 ml/kg/24 hrs

Medium Dogs

5%

Body weight kg	Deficit ml	Maintenance ml	Drops/min***	ml/hour in 24 hours*
11	550	550	15.2	45.8
12	600	600	16.6	50
13	650	650	18	54.1
14	700	700	19.4	58.3
15	750	750	20.8	62.5
16	800	800	22.2	66.6
17	850	850	23.6	70.8
18	900	900	25	75
19	950	950	26.3	79.1
20	1000	1000	27.7	83.3
21	1050	1050	29.1	87.5
22	1100	1100	30.5	91.6
23	1150	1150	31.9	95.8
24	1200	1200	33.3	100
25	1250	1250	34.7	104.1

10%

Body weight kg	Deficit ml	Maintenance ml	Drops/min***	ml/hour in 24 hours*
11	1100	550	22.9	68.7
12	1200	600	25	75
13	1300	650	27	81.2
14	1400	700	29.1	87.5
15	1500	750	31.2	93.7
16	1600	800	33.3	100
17	1700	850	35.4	106.2
18	1800	900	37.5	112.5
19	1900	950	39.5	118.7
20	2000	1000	41.6	125
21	2100	1050	43.7	131.2
22	2200	1100	45.8	137.5
23	2300	1150	47.9	143.7
24	2400	1200	50	150
25	2500	1250	52	156.2

Large Dogs

5%

Body weight kg	Deficit ml	Maintenance ml	Drops/min***	ml/hour in 24 hours*
26	1300	1040	32.5	97.5
28	1400	1120	35	105
30	1500	1200	37.5	112.5
32	1600	1280	40	120
34	1700	1360	42.5	127.5
36	1800	1440	45	135
38	1900	1520	47.5	142.5
40	2000	1600	50	150
42	2100	1680	52.5	157.5
44	2200	1760	55	165
46	2300	1840	57.5	172.5
48	2400	1920	60	180
50	2500	2000	62.5	187.5

10%

Body weight kg	Deficit ml	Maintenance ml	Drops/min***	ml/hour in 24 hours*
26	2600	1040	50.5	151.6
28	2800	1120	54.4	163.3
30	3000	1200	58.3	175
32	3200	1280	62.2	186.6
34	3400	1360	66.1	198.3
36	3600	1440	70	210
38	3800	1520	73.8	221.6
40	4000	1600	77.7	233.3
42	4200	1680	81.6	245
44	4400	1760	85.5	256.6
46	4600	1840	89.4	268.3
48	4800	1920	93.3	280
50	5000	2000	97.2	291.6

* This value should never be rounded up, and this will lead to over infusion.

** Using a 60 drops/ml giving set.

*** Using a 20 drops/ml giving set.

Types Of Fluid

Fluids can be broken down into 2 types:

1. Crystalloid
2. Colloids

Crystalloids

Crystalloids are the 'saline' type fluids. They contain varying amounts of sodium. They also contain water, electrolytes and non-electrolyte ingredients.

Crystalloids are broken down into two groups:

1. Replacement fluids
2. Maintenance fluids

Replacement fluids

Replacement fluids have the same electrolyte makeup as plasma; this means that large quantities can be given quickly. The problem is that these fluids will equilibrate rapidly with interstitial fluids. This means that only 15 – 25% will be left in the extra vascular space after 1 hour of administration. Replacement fluids are indicated in cases where fluid and electrolytes have been lost for example in vomiting, diarrhoea, polyuria, effusions and haemorrhage.

Common replacements fluids

Normal saline (0.9% sodium chloride), Ringers solution, Hartmanns (Vetivex 11) and hypertonic saline (7.2% sodium chloride).

Maintenance fluids

Maintenance fluids are used once an animal is rehydrated and ongoing losses are being controlled. It is then necessary to change the animal to a fluid with lower sodium content; this lessens the risk of hypernatraemia (elevated blood sodium level) and peripheral oedema (swelling of tissues in the legs). Low sodium fluids are useful when managing patients with congestive heart failure (CHF) and renal disease. Low sodium fluids should be used for patients that have CHF and are dehydrated. Patients with renal disease are less able to filter a high sodium fluid and excrete large amounts of potassium, in such cases a low sodium fluid would be more suitable.

Common maintenance fluids

Dextrose Saline, 0.18% saline in 4% dextrose (Vetivex18 or Aquapharm No.18).

Hypertonic Saline: Sodium Chloride 7.2%

Types of Crystalloids

This is a special type of replacement fluid used for acute fluid resuscitation only. Giving a fluid with a high sodium level (approx 8 times normal plasma level) causes any free water to move into the intravascular space from the interstitial and intracellular space. This happens rapidly, within 5 minutes. The down side to this type of fluid is as soon as equilibrium is reached, the fluid returns to the interstitial space, and its effects are short lived.

The recommended dose for hypertonic saline is 4 – 8 ml/kg and a rate of 60 ml/kg/per hour should not be exceeded. It should be delivered over 5 minutes. If administration is too quick then bronchoconstriction (narrowing of the respiratory tract) and bradycardia (slow heart rate) can occur. Its effect is short lived at approximately 30 – 60 minutes.

Hypertonic Saline has the advantage in that a small dose can be administered quickly, and is more cost effective than colloids, especially when dealing with giant dogs. Hartmanns or Saline (0.9% sodium chloride) should be started as a follow up treatment, and water should be offered unless water is contraindicated due to the animals' condition. Hypertonic Saline (7.2% sodium chloride) is useful where hypovolaemia is not caused by normal bleeding such as in Gastric Dilation Volvulus (GDV). Hypertonic saline should not be used in animals that are dehydrated.

Hartmanns Solution

Hartmanns solution is a solution containing sodium chloride, sodium lactate, and phosphates of calcium and potassium. It is isotonic, so can be used intravenously and also via subcutaneous injection. It is used to replace body fluid and mineral salts that may be lost for a variety of medical reasons. Hartmanns solution is ideal for use in patients with hypovolaemia. Care should be taken with patients' with / or potential for cerebral oedema (brain swelling) as fluid will move from the extra cellular space into the intra cellular space because Hartmanns is slightly hypotonic. Hartmanns Solution should also be used in patients with metabolic acidosis, gastroenteritis, renal insufficiency, diabetic ketoacidosis, GDV, Pyometra and shock. Patients with alkalosis should not be given Hartmanns due to the bicarbonate; they should be given normal saline instead. Metronidazole should never be added to a bag of Hartmanns due to a reaction causing crystallisation of particles.

Ringers' Solution

Ringers solution is a solution that contains potassium, calcium and has a sodium level of 147 mmol/l, it is isotonic in cats and dogs but the chloride level is much higher than that of the plasma. Ringers has no source of bicarbonate and high levels of chloride and therefore it is not as effective at treating metabolic acidosis, in cases of metabolic acidosis Hartmanns should be used instead. Ringers may be used to correct volume deficit, which has been a result of shock or gastrointestinal disease. Ringers is especially useful in cases of hypokalaemia resulting from prolonged vomiting.

Both Hartmanns and Ringers should not be infused through the same catheter as blood products, the combination can cause thrombi to form which can then pass into the blood stream.

continued...

Normal Saline – 0.9% sodium chloride

This is the oldest type of infusion solution, containing only sodium chloride and water. It does not contain the same balance of electrolytes as plasma and so should not be used as a universal fluid. Patients that are suffering from metabolic acidosis should be placed on normal saline as it lowers the blood plasma bicarbonate. Animals suffering from vomiting and animals on Furosemide BP (Vet) may benefit from being on normal saline. Normal saline can be used for animals with hyperkalemia, as it does not contain any potassium, but Hartmanns is probably a better choice as the potassium level in Hartmanns is quite low, and as Hartmanns contains bicarbonate it may be beneficial as hyperkalaemic patients are often acidotic as well. 0.9% sodium chloride can be useful in hypercalcaemic patients because the high sodium levels compete with calcium for re-absorption by the kidney and the calcium is excreted in the urine.



Colloids

Colloids are given to patients when crystalloids will not be sufficient to support the circulation. Colloids are solutions that contain large molecules, and these remain in the circulation longer, whereas crystalloids are distributed around the body rapidly. The colloids entering the circulation and staying there cause the plasma volume to expand. This is useful in conditions such as hypovolaemia, hypotension and shock. The most commonly used colloids are synthetic, there are 3 types: Dextrans, Gelatins and Hydroxyethyl starches. The advantages of colloids are; they have a long shelf life, are low cost in comparison to plasma products and are less immunogenic (less likely to cause allergic reaction).

Dextrans 40 and Gelatins like heamaccel® and Gelofusin® have relatively low molecular weights (30 – 40kD) and have a limited duration of effectiveness of less than 6 hours. Dextran 70 has a higher molecular weight at 70kD it works for longer at 6 – 12 hours.

There are 3 types of Hydroxyethyl starch with various molecular weights, all larger than the Gelatins and Dextrans, this means that their duration of effect will be longer depending on the clinical condition of the animal. The duration of effect is dependent on the molecular weight but the number of particles in the solution causes the volume expansion. In hypovolaemic dogs and cats the dose of colloid should be 5 ml/kg over 10 – 30 minutes. The dose may be repeated but care should be taken with cats as volume overload can cause pulmonary oedema. Therefore as a general rule do not infuse more than 20 ml/kg in 24 hours.

All colloids carry a risk of anaphylactic reaction; this is because they are made from ingredients not naturally made in the animal, in particular, gelatins as they are made from a purified bovine source. Colloids also have the potential to affect renal function so an animal on colloids should have its urine output monitored and also watched for clinical signs of renal insufficiency.

Common colloids

Gelofusin®, Heamaccel®, Voluven®.

Types of Colloids

Gelatins

These come from bovine collagen that has been treated to make an albumin type molecule. These products have been around for many years, the most commonly ones used are Haemaccel® and Gelofusin®, which are the only UK veterinary licensed products. Gelatins are used for the treatment of hypovolaemic shock that has occurred due to bleeding, trauma, sepsis, burns, peritonitis and surgery. Gelatins are also useful for short-term blood loss that may result during general anaesthesia. Gelatins have little effect on clotting but are the most likely group of colloids to cause anaphylactic reaction.

Dextrans

Dextrans are made from glucose polymers that are created in the lab, and do not come from a particular species. They are available in two molecular weights Dextrans 40 – 40kD and Dextrans 70 – 70kD. They have a short-term effect on hypovolaemia, lasting 3 – 12 hours, Dextrans 70 will last longer than Dextrans 40. There is a slight risk of anaphylactic reaction, and Dextrans 40 has been reported to cause renal failure in some human cases. Dextrans are more commonly used in large animal medicine.

Hydroxyethyl Starches (HES), Voluven®

These types of colloids are made from starch. 'HES' is particularly useful in cases of prolonged hypovolaemia, in cases such as sepsis and severe inflammation. These animals often have leaky capillaries and due to the molecular weight of the HES the colloid remains in the blood vessels for longer, and therefore maintains circulatory volume for longer. HES can interfere with normal platelet function (shown during lab tests) and may cause an increase in clotting times. As with any colloid, renal function should be monitored as the filtration of such large molecules puts a large burden on the kidneys. Care should be taken when using colloids in patients with pre-existing renal conditions.

Intravenous cannulisation

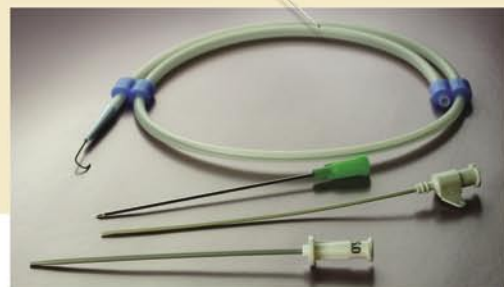
Over-the-needle cannulae (Anicath™ IV Cannulae, Anicath™ Non Winged, Anicath Xtra™)

These are the most commonly used cannulas in veterinary practice to administer intravenous fluids. They are generally short, being 25 – 50 mm long depending on the size of animal they are being placed into. They consist of polyurethane or Teflon outer lumen with a metal stylet running through the cannula, this stylet / needle is only used to introduce the cannula into the vein, and then it is removed and discarded. They can be winged or wingless. They can be left in place for up to 72 hours and are usually placed in the cephalic, saphenous (lateral or medial), jugular and marginal ear vein.



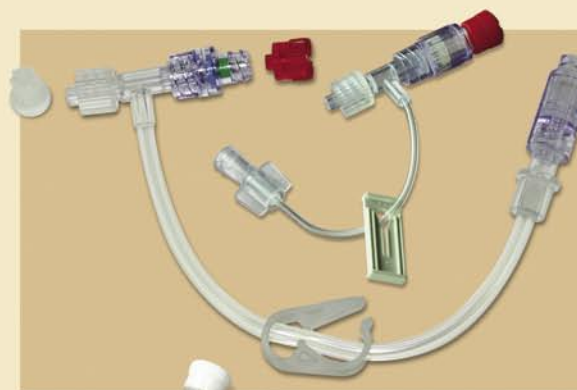
Butterfly cannulae (Anicath™ IV Wing Set)

These consist of a metal needle and plastic wings and sometimes a short piece of tubing with an injection port is also attached. The wings help secure the cannulae in place. This type of cannulae often becomes dislodged or falls out, and can cause phlebitis because the needle is rigid.



Through-the-needle cannulae

These are usually longer than the over-the-needle cannulae and are most commonly used to cannularise the jugular vein. They must be placed in a sterile manner using sterile gloves and drapes. They can be left in place for up to 7 days, and are often used to measure central venous pressure. They can also be used to administer fluids or drugs and collection of blood samples. The disadvantage of these cannulas is that they are much more expensive and are more difficult to place.



T-connectors (Aniset™-T-Connectors)

T-connectors are placed between the cannula and the giving set. They usually include an injection port and a plastic clamp. This allows the cannula to be flushed with heparinized saline without removing the giving set and can also be used to administer intravenous drugs. One of the main benefits of a t-connector is that it allows you to disconnect the drip from your patient in order to examine or exercise your patient.



Burettes (Aniflow™ Burette)

This is a large graduated chamber and is used when a very small amount of fluid needs to be given, i.e. in the case of a small mammal such as rabbits, where a certain amount of drug has to be given. The burette is usually placed between the fluid bag and the drip chamber. They can come as separate pieces of infusion equipment attached to a standard giving set or are part of the infusion system. The burette is filled by opening the roller clamp between the fluid bag and burette allowing the fluid to fill the burette to the desired amount or up to 150ml. The clamp below the burette should be closed, to prevent air entering the rest of the giving set.



Extension sets (Anitube-S™, Ani-Y-Port™, Aniset Standard Flow Extension™, Aniset High Flow Extension™)

These are additional lengths of tubing that are placed between the cannula or Aniset™ T-connector and the original giving set thus making it longer. This is useful for medium to large dogs who require bigger kennels and therefore need more tubing to be able to move around their kennels. The extension sets can either be straight or coiled.

continued...

Fluid Therapy Equipment

10

Giving sets (Aniset™)

This is the link between the fluid and the animal cannula. A giving set is made up of a long piece of tubing with a drip chamber and spike at one end and a leuc lock connector at the other. A flow regulator in the form of a dial (in the case of dial-a-flow giving sets) or roller clamps are positioned below the drip chamber, which allows the fluid rate to be controlled. There may be one or more injection ports to allow for the administration of drugs or to facilitate the removal of air bubbles from the set.

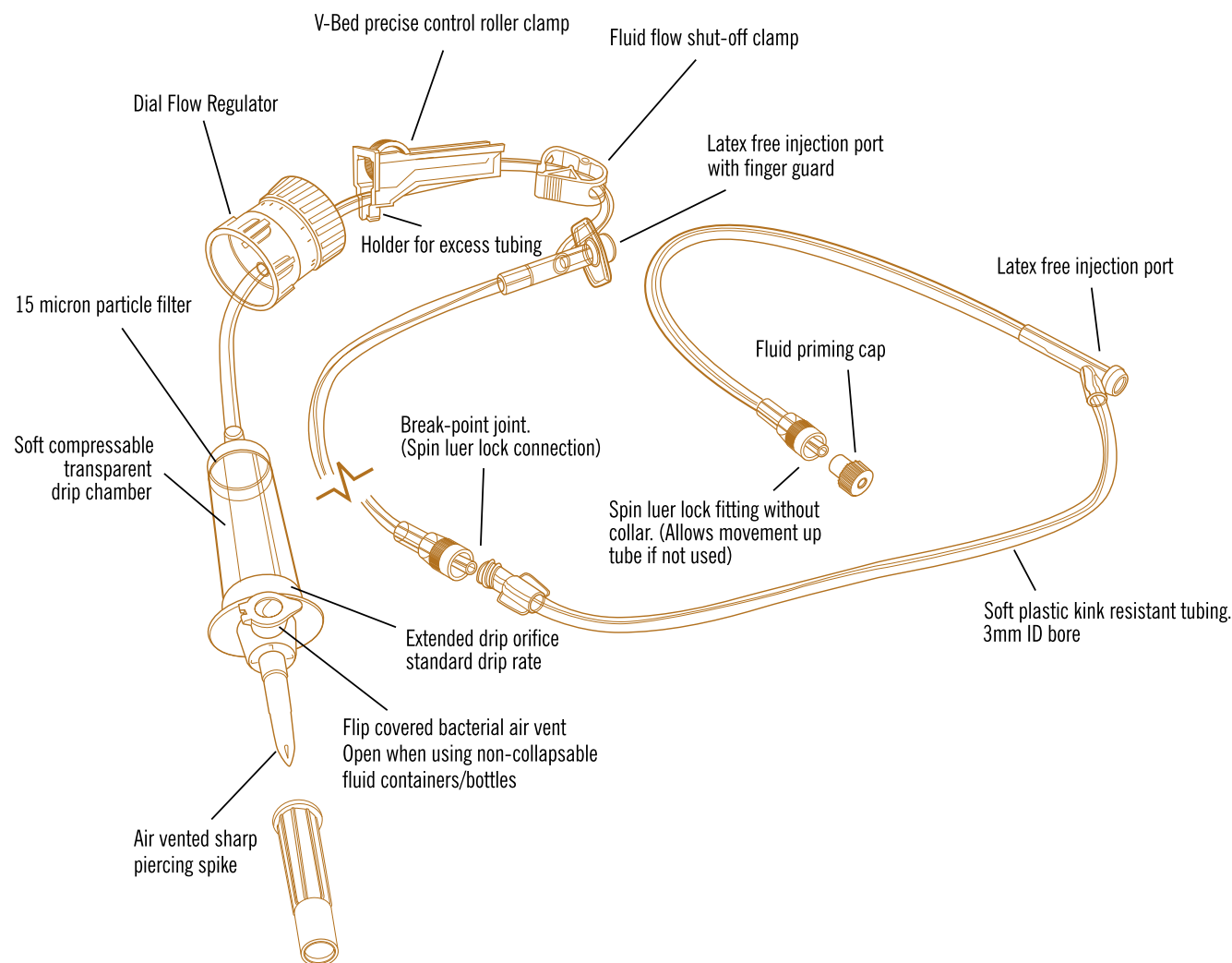
There are 4 types of giving set:



1. Standard sets that deliver 20 drops/ml



2. Paediatric sets that deliver 60 drops/ml



3. Aniset™ Dial-A-Flow sets, can deliver 20 drops/ml and 60 drops/ml. The advantage of the dial a flow sets over standard giving sets is that the dial allows you to control the rate that the fluid is given, usually in ml/hr. (Like an infusion pump but without the expense of buying one.) This lessens the chance of over infusion and potentially serious pulmonary oedema.



4. Giving sets that are used with volumetric infusion pumps. These are giving sets that have been specially adapted to be used with certain infusion pumps. It is recommended that you check with the manufacturer of the infusion pump to ensure it is compatible with your set.

How To Place A Cephalic Intravenous Cannula

11

Equipment



The saphenous vein can also be used where access to the cephalic is not possible, i.e. if the animal has already had cannula placement in both legs or if, there are wounds on the legs etc. However, you should consider the possibility of infection from contamination if the animal is suffering from diarrhoea.

1. Over-the-needle intravenous cannulae (Anicath Xtra™, Anicath™ IV Cannulae, Anicath™ Non Winged)
2. Aniset™ t-connector
3. Clippers
4. Heparinized saline (0.1ml 1:1000 heparin in 100ml 0.9% saline)
5. Chlorhexidine surgical scrub (diluted 1:16)
6. Surgical spirit
7. Millswabs™, Zorbs™ Gauze Balls, Millsoft™ cotton wool
8. Adhesive tape, ideally hypoallergenic (Anifilm™, Anipore™, Anisilk™, E-band™)
9. Orthoband™ bandage
10. Wrapz™ or Co-form™ bandage

Optional:

- Local anaesthetic cream (EMLA Cream)
- Sterile gloves
- Scalpel blade (size 11)

INFUSION ACCESSORIES



continued...

How To Place A Cephalic Intravenous Cannula

11

Placing your cannulae

Ideally, an aseptic technique using sterile gloves should be adopted, but in reality this is rarely done. If you are not using surgical gloves then your hands should be washed thoroughly.

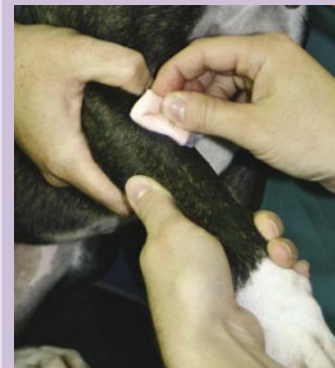


1. Gather the equipment and flush the Aniset™ t-connector with heparinized saline.



2. Get an assistant to restrain the patient following correct handling and restraining techniques.

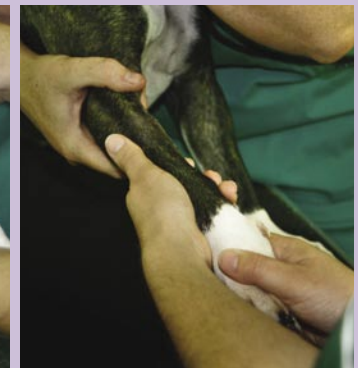
3. Clip the hair over the vein to be catheterised and also around the entire leg. Identify the vein visually and by feel.



4. Clean the area with Chlorhexidine (dilute 1:16).



5. Wipe the area with a Millswab™ wetted with surgical spirit.



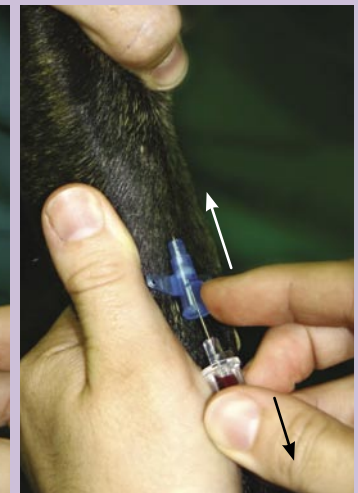
6. Ask your assistant to raise the vein. It is important your assistant does not twist the vein as this may lead to problems with cannula placement once your assistant releases the vein.



7. Wash your hands again, the surgical area should now not be touched. Introduce the cannula into the vein. The 'cutdown' technique should only be used when visualisation of the vein is impossible.



8. Once blood returns into the cannula hub, hold the stylet still and advance the cannula into the vein.



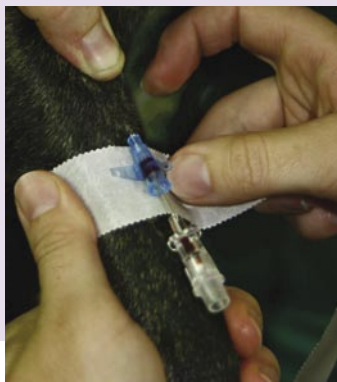
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How To Place A Cephalic Intravenous Cannula

11



9. Ask your assistant to stop raising the vein and press over the vein a few centimetres above the cannula entry site, this stops the blood flowing out of the cannula when the stylet is completely removed.



10. Use a length of tape to secure the cannula. Start by placing the tape underneath the cannula directly on the patients' skin and then wrap around the leg and over the cannula.



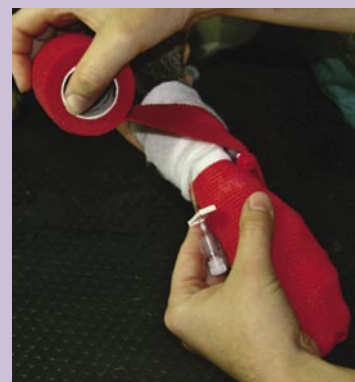
11. Remove the stylet and attach the Aniset™ t-connector. Secure the Aniset™ t-connector in the same manner as the cannula using tape.



12. Flush the cannula with some more heparinized saline to check placement. There should be very little resistance when injecting and there should be no swelling around the cannula.



13. The animal can now either have its leg bandaged using the padding bandage and cohesive bandage to protect the cannula and Aniset™ t-connector or it can be attached to a giving set to receive fluids, and then bandaged.

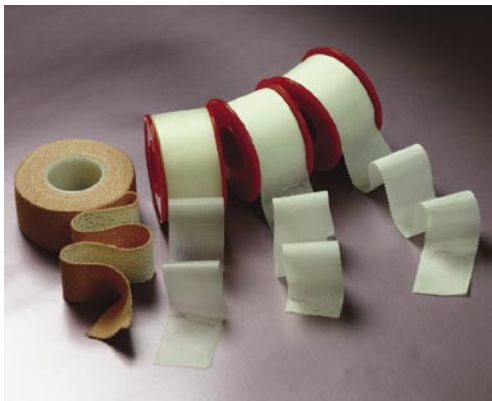
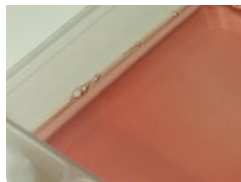


How To Place A Cannula In The Marginal Ear Vein Of A Rabbit

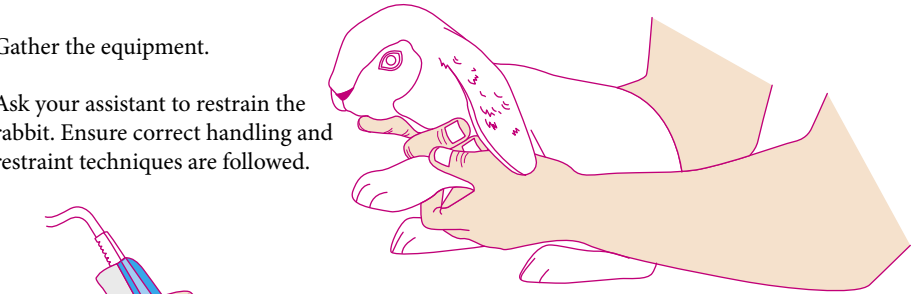
12

Equipment

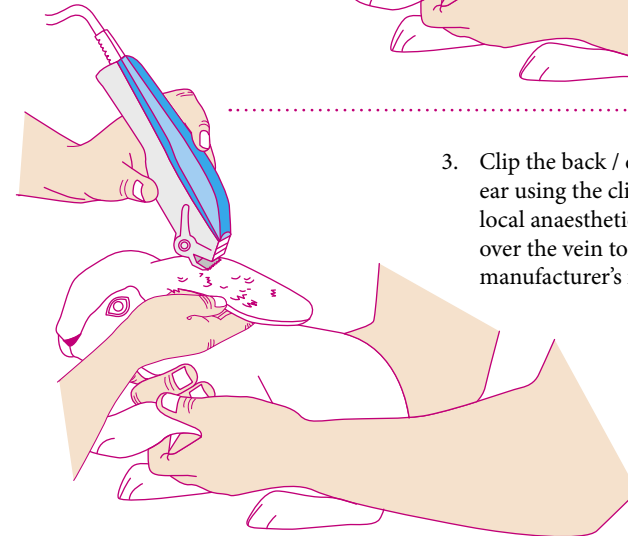
1. Local anaesthetic cream (EMLA cream)
2. 26g or 24g Over-the-needle Cannula (Anicath™)
3. Clippers
4. Heparinized saline (0.1ml 1:1000 heparin in 100ml 0.9% saline)
5. Chlorhexidine surgical scrub (diluted 1:16)
6. Surgical scrub
7. Swabs / cotton wool (Millswabs™ / Millsoft™)
8. Adhesive tape, ideally hypoallergenic (Anifilm™, Anipore™, Anisilk™, E-band™)
9. Orthoband™
10. Cohesive bandage (Wrapz™ or Co-form)



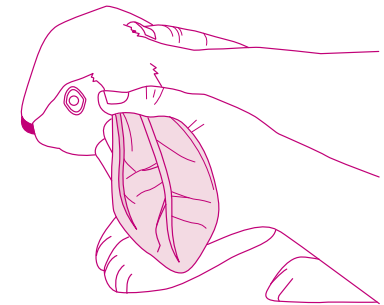
1. Gather the equipment.
2. Ask your assistant to restrain the rabbit. Ensure correct handling and restraint techniques are followed.



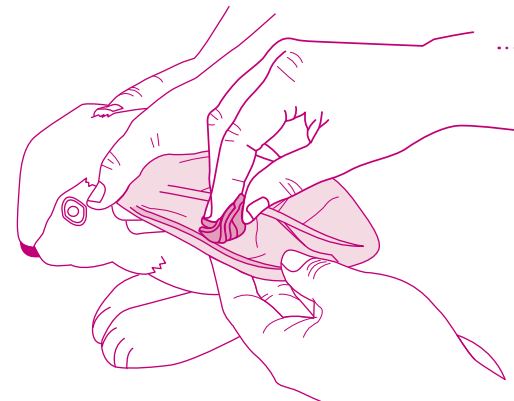
3. Clip the back / dorsal aspect of the ear using the clippers and apply some local anaesthetic cream to the area over the vein to be used and leave as manufacturer's instructions.



4. Ask your assistant to raise the vein by applying pressure to the base of the ear. The marginal ear veins are the veins that run around the outside of the ear. The vein that runs up the middle should never be used.



5. Using dilute chlorhexidine scrub, remove any remaining local anaesthetic cream and clean the area.

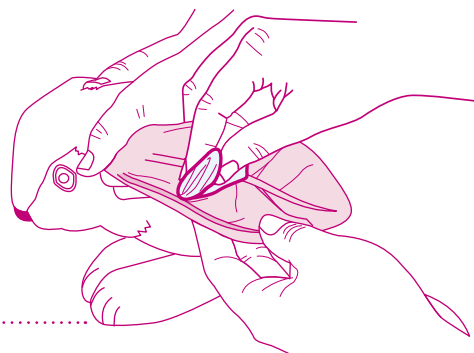


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How To Place A Cannulae In The Marginal Ear Vein Of A Rabbit

12

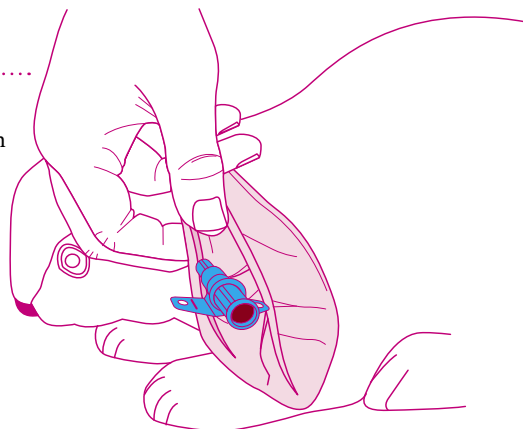
6. Wipe with surgical spirit.



7. Identify the vein and insert the cannula through the skin into the vein. The marginal ear vein is very shallow, extremely delicate and care should be taken not to perforate the other side of the vein and 'blow' the vein, causing an aural haematoma.

8. Once blood returns into the hub of the cannula, gently advance the cannula into the vein whilst holding the stylet still.

9. Ask your assistant to stop raising the vein and apply gentle pressure to the area just below the cannulae entry site to prevent blood escaping the cannula when the stylet is removed.

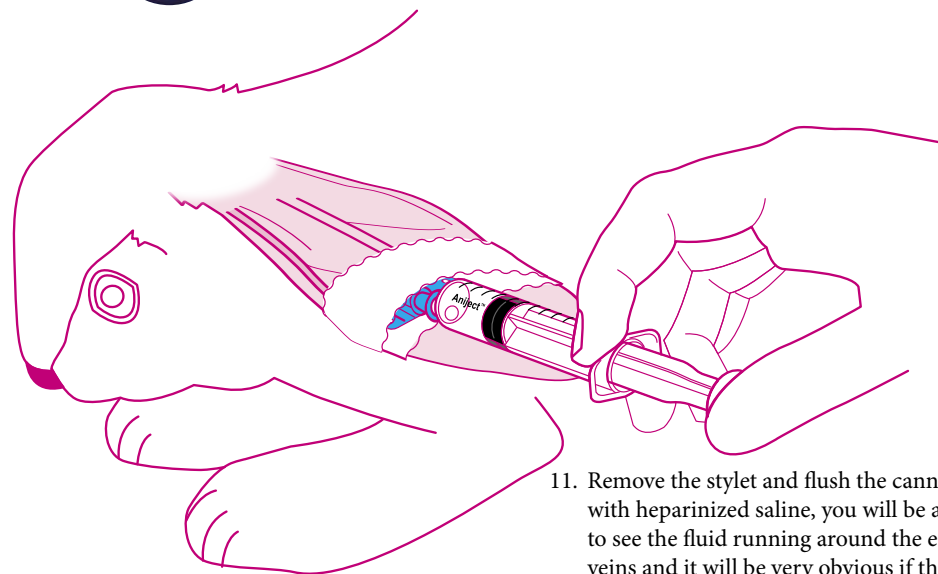


10. Apply tape around the ear, starting underneath the cannula and coming around the ear and back over the cannula. Do not pull the tape too tight, the ear should stay a natural shape.



TOP TIP

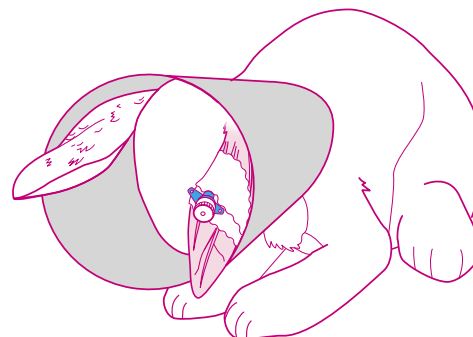
Use a small, folded piece of Orthoband™ on the inner aspect of the pinna to help maintain natural pinna shape.



11. Remove the stylet and flush the cannula with heparinized saline, you will be able to see the fluid running around the ear veins and it will be very obvious if the cannula is not in the vein.

12. An injection bung or giving set can now be attached to the cannula. More tape will be needed to secure a giving set to the ear to prevent it pulling the cannula out. It is the author's experience that a little trial and error is needed to find the best technique to securing the giving set to the ear.

13. An Elizabethan collar should then be put on the rabbit to prevent it scratching out the cannula and giving set.



How To Set Up A Drip

13

Equipment

1. Giving set (Aniset™)
2. Heparinized saline (0.1ml 1:1000 heparin in 100ml 0.9% saline)
3. Intravenous fluids



1. Collect together equipment and flush Aniset™ t-connector with heparinized saline.



2. Ensure that the cannula has been placed and is patent. Attach Aniset™ t-connector to patient if one is not already present. (See page 35)
3. Check the packaging of the giving set to ensure that it is not damaged and / or out of date. Also check that the fluids are in date, are not discoloured or have any sediment present.



4. Remove the giving set from the package and close the flow regulating clamp.



5. Remove the cap from the spike on the giving set and push it in to the port of the fluid bag.



6. Half fill the drip chamber by squeezing it.



7. Open the flow regulating clamp slowly to allow the fluid to flow through the giving set until it is full, this is known as 'priming'. Fully close the flow regulator.



8. Check that there are no air bubbles in the line.
9. Suspend the fluid bag from a drip stand.
10. Connect the end of the giving set to the t-connector, open the clamp on the Aniset™ t-connector. (See page 37)
11. Open the flow regulator so the fluid is delivered at the correct infusion rate. (See page 18-20)

Monitoring The Fluid Therapy Patient

It is very important that once a patient is receiving fluid therapy that it is monitored very closely and vital signs and any comments are recorded and then reported to the veterinary surgeon.

Parameters that should be monitored at least twice daily while the patient is receiving fluid therapy are:

Cardiovascular system

- Pulse-rate, rhythm and strength.
- Mucus membrane colour.
- Capillary refill time.
- Check for any jugular distension.
- Auscultation of the chest
 - cardiac arrhythmias.

Respiratory system

- Respiratory rate and depth.
- Chest auscultation
 - pulmonary oedema.

Body Temperature

Normal body temperature ranges:

Cat	Dog	Rabbit
38.0°C – 38.5°C	38.3°C – 38.7 °C	38.5°C – 40°C

Urine output-normal

- 1 – 2 ml/kg/hr

Urine specific gravity

Normal urine SG ranges:

Cat	Dog	Rabbit
1.020 – 1.060	1.015 – 1.045	1.003 – 1.036

General checks

- Peripheral oedema – hocks, over the dorsum. Oedema of the scruff is quite easy to spot.
- Chemosis – the mucus membranes of the eye become puffy.
- Skin turgor – rate at which the skin rapidly returns to its normal position.
- Bodyweight

Normal blood serum levels of Na⁺, K, Cl⁻, and Bicarbonate.

It is highly advisable you should record what the animal takes in (food, water, iv fluids) and record what comes out of the animal. (urine, faeces, vomit, diarrhoea). The best way to do this is via a hospital sheet or fluid therapy monitoring sheet page 46, then the veterinary surgeon can build up an entire picture of how the animal is progressing.

Monitoring what is going in

All food and water should be measured before it is given to the patient. This may mean finding out exactly how much water is in a tin of dog food. It is then easy to measure what is left after an allocated amount of time and this can tell us what the animal has taken in. You should also include what the animal receives in the way of intravenous fluids.

Monitoring what is coming out

Estimate volume of vomit and diarrhoea. The easiest way to monitor urine output is via a urinary catheter, and this is fine if you have a recumbant patient, that requires an indwelling urinary catheter. If your patient is more stable then they may not have one in situ. In this case you just need to weigh the wet bedding against the dry bedding. Weigh all the animal bedding before it goes into the kennel, use non-absorbent cat litter, nappies or cage liners (MP Underpads™). After an allotted time weigh it all again. You can safely assume that 1ml of urine weighs 1g. The increase in weight will tell you how much urine the animal has passed.

The 'ins' and 'outs' should be measured and recorded every 4 hours. Ideally, what comes out should be about 10% smaller than what went in.

If the animal is excreting more than is being given, the fluid rate should be increased. If the animal is taking in a lot more than is being excreted then this should be investigated, and the intravenous fluid rate decreased. It may be that the animal has not been on intravenous fluids long enough and you are still correcting the dehydration. Therefore, in this situation it is reasonable to expect the animal to still be retaining some fluid. This should be checked again in a couple of hours and your findings recorded.

Fluid Overload

This is when the patient receives too much fluid. This is quite common in small mammals, cats and small dogs, which is why fluids must be carefully monitored. Fluid overload can lead to peripheral oedema, which can lead more seriously to pulmonary oedema and can lead to respiratory distress and death.

Excessive administration of colloids can lead to right sided heart failure, and congestive heart failure. Crystalloid overload, in the early stages causes diuresis – the animal may produce lots of very dilute urine resulting in an increased risk of peripheral oedema and then pulmonary oedema, and death.

15

DATE PLACED:

FLUID TYPE	AMOUNT TO BE INFUSED TOTAL	MLS PER HOUR	DRIP RATE

DIET: WATER:

[illegible][illegible][illegible]

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You can also cause fluid overload in patients with chronic anaemia by giving them too much blood. They are essentially normovolemic.



Administering a blood transfusion

On rare occasions it may be necessary to administer blood to a patient. The description below explains the correct use of the Aniset™ Transfusion blood giving set, and also contains the correct calculation for the administration of blood.

We have included a quick reference table to help you decide how much blood you give initially, and how quickly to give the remaining blood (assuming that you haven't had any problems.)

- Close the roller clamp.
- Remove protector from piercing spike and insert into bag of blood that is to be transfused.
- Gently squeeze the drip chamber until it is half full of blood.
- Whilst holding the end of the Aniset™ Transfusion set, slowly open the roller clamp until the blood has reached the priming cap. Close the roller clamp.
- Remove priming cap and attach the Aniset™ Transfusion set to the indwelling IV cannula of the transfusion recipient.
- Blood should be initially administered at a rate of 1 ml/kg/hr, and the recipient observed for signs of transfusion reaction.
- After 30 minutes and if no problems have been observed, increase the infusion rate so that the remaining blood is given over 4 hours.



In cases where severe blood loss has occurred it may be necessary to infuse the blood more rapidly to save the animals life.

Blood Transfusion Chart

Give 1 ml/kg/hr for 30 mins

If after 30 mins there is no sign of transfusion reaction give the remaining blood over 4 hours.

Cats – based on a donation of 40ml

Recipient weight	Initial rate	Remaining blood after 30min blood	Rate to give remaining
2 kg	1 ml/hr	39 ml	9.7 ml/hr
3 kg	1.5 ml/hr	38.5 ml	9.6 ml/hr
4 kg	2 ml/hr	38 ml	9.5 ml/hr
5 kg	2.5 ml/hr	37.5 ml	9.3 ml/hr
6 kg	3 ml/hr	37 ml	9.2 ml/hr
7 kg	3.5 ml/hr	36.5 ml	9.1 ml/hr

Dogs – based on 500ml donation

Recipient weight	Initial rate for 30 min	Remaining blood after 30min blood	Rate to give remaining
4 kg	2 ml/hr	498 ml	124.5 ml/hr
6 kg	3 ml/hr	497 ml	124.2 ml/hr
8 kg	4 ml/hr	496 ml	124 ml/hr
10 kg	5 ml/hr	495 ml	123.7 ml/hr
12 kg	6 ml/hr	494 ml	123.5 ml/hr
14 kg	7 ml/hr	493 ml	123.2 ml/hr
16 kg	8 ml/hr	492 ml	123 ml/hr
18 kg	9 ml/hr	491 ml	122.7 ml/hr
20 kg	10 ml/hr	490 ml	122.5 ml/hr
22 kg	11 ml/hr	489 ml	122.2 ml/hr
24 kg	12 ml/hr	488 ml	122 ml/hr
26 kg	13 ml/hr	487 ml	121.7 ml/hr
28 kg	14 ml/hr	486 ml	121.5 ml/hr
30 kg	15 ml/hr	485 ml	121.2 ml/hr
32 kg	16 ml/hr	484 ml	121 ml/hr
34 kg	17 ml/hr	483 ml	120.7 ml/hr
36 kg	18 ml/hr	482 ml	120.5 ml/hr
38 kg	19 ml/hr	481 ml	120.2 ml/hr
40 kg	20 ml/hr	480 ml	120 ml/hr

Glossary And Prefixes

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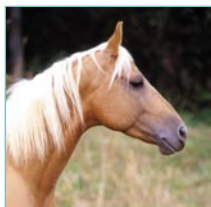
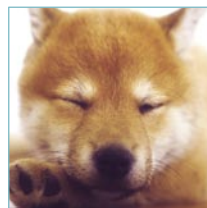
Acute	– Sudden onset			a scalpel blade) to facilitate intravenous cannulae placement
Albumin	– A protein in the blood responsible for the maintenance of osmotic (water) pressure in the blood; also binds (attaches) to large molecules in the blood and serves to transport them; produced by the liver; also called 'serum albumin.'	Distal		– Situated away from the centre of the body
		EDTA		– Ethylenediaminetetraacetic acid, an anticoagulant
		Equilibrium		– A state of balance
		Extracellular fluid		– Fluid found outside the cells
		GDV		– Gastric dilation volvulus, condition where the stomach twist's and occludes and causes a greatly distended abdomen
Anaemia	– A condition in which the number of red blood cells present in the blood is lower than normal			
Anaphylactic	– Allergic reaction to something	Globulin		– Plasma protein, large molecules that can not pass out of the circulation to the tissues, and so they help to maintain the osmotic pressure of the blood
Aspiration pneumonia	– Pneumonia (disease of the lungs characterized by inflammation and consolidation of lung tissue) caused by inhalation of fluid into the lungs			
		Haemoglobin		– A protein inside of red blood cells, responsible for the binding and transport of oxygen to the body tissues (Hb)
Bradycardia	– Slow heart rate			
Bronchoconstriction	– Narrowing of the respiratory tract			
Cardiac arrhythmia	– A variation from normal heart rhythm			
Caudal	– A directional term used to refer to an area more toward the cauda, or tail region; opposite of cranial	Hepatic		– Relating to the liver
		Hyperglobulinaemia		– An abnormally large amount of globulins in the circulating blood plasma
		Hyperkaleamia		– High blood potassium
Central venous pressure	– Describes the pressure of blood in the thoracic vena cava, near the right atrium of the heart. CVP reflects the amount of blood returning to the heart and the ability of the heart to pump the blood into the arterial system	Hypernatraemia		– Elevated blood sodium level
		Hypertonic saline		– Has a higher concentration of sodium (approximately 8 times the normal plasma concentration of sodium) which causes free water to migrate into the intravascular space from the interstitial and intracellular spaces
Cephalic	– Large vein in the foreleg			
Chronic	– Long term i.e. not sudden onset	Hyperventilating		– An increase in the rate and/or depth of respiration
Cl-	– Chloride	Hypocalcaemia		– Low blood calcium
Congestive heart failure	– Inability of the heart to supply sufficient blood flow to meet the needs of the body	Hypoproteinaemia		– A condition where there is an abnormally low level of protein in the blood
Cranial	– Towards the head of the body	Hypotension		– Low blood pressure
Crea	– Breakdown product of creatinine phosphate in the muscle, excreted by the kidneys	Hypoventilating		– A decrease in the rate and / or depth of respiration
		Hypovolaemia		– Low circulating blood volume
CRT	– Capillary refill time, the time it takes for the blood to return to the tissues after pressure has been applied	Immunogenic		– Less likely to cause allergic reaction
		Interstitial		– Space that surrounds the cells of a given tissue
Crystalloid	– Saline type fluid	Intracellular fluid		– Fluid found inside the cells
Cut down technique	– Method where you incise the skin over the vein (with	Intraosseous		– Inside the bone
		Intraperitoneal		– Inside the peritoneal cavity
		Intravascular		– Within the blood vessel

Jugular distention	- Protruding of the jugular vein		or medial (on the inner aspect of the hind leg)
K+	- Potassium	Sepsis	- The presence of toxins in the blood or other tissues; the toxins are produced by bacteria or other microorganisms
Medial	- Towards the midline	Splenic contraction	- When the spleen contracts, it injects large numbers of red blood cells into circulation, thereby increasing the relative ratio of red cells to fluid in the blood
MM	- Specialized membrane which covers various passages and cavities exposed to the air such as the mouth, nose, inner portion of the eyelids, vagina. Examination of the mucous membranes can provide important information: if they are dry, the animal is likely dehydrated; pale, and the animal may be anaemic or in shock; yellow, and the animal is said to be jaundiced due to accumulation of waste products which should be eliminated by the liver	Subcutaneous	- Under the skin; often called 'sub Q'
Na+	- Sodium	Thrombi	- A blood clot
Oedema	- Accumulation of fluid in one or more body cavities or tissues	Umbilicus	- The area of the body where the umbilical cord is attached; the belly button
Oliguria	- Reduced excretion of urine	Urea	- Waste product of protein metabolism that is removed from the body by the kidneys
PCV	- Packed cell volume. A laboratory test to monitor the relative number of red blood cells present in the blood	Urethral obstruction	- Obstruction of the urethra with calculi/sediment causing the animal to be unable to urinate
Peripheral	- Near the surface or outside of; external	Urine specific gravity	- Urine specific gravity is a laboratory test that measures the concentration of all chemical particles in the urine
Peritonitis	- Inflammation of the peritoneum (lining of the abdomen)		
Plasma	- The fluid part of the blood		
Polycythemia	- Refers to an increase in either the number or concentration of red blood cells in circulation. Polycythemia is also called erythrocytosis, because red blood cells are medically referred to as "erythrocytes"		
Polyuria	- Excessive urination		
Proximal	- The end of limb nearest the body		
Pulmonary	- Pertaining to the lungs		
Pyometra	- Pus filled uterus		
Renal	- Relating to the kidneys		
Right sided heart failure	- The right ventricle loses its pumping function, and blood may back up into other areas of the body, producing congestion. Congestion affects the liver, the gastrointestinal tract, and the limbs. In addition, the right ventricle may be unable to pump blood efficiently to the lungs and to the left ventricle		
Saphenous	- Large superficial veins in the hind limb, can be lateral (outer aspect of the hind leg)		





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